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## SHORT PULSE KINETICS EXPERIMENTS

### 1. KrF* and HgX* (X = Br, I) Kinetics

Initial fast-photolysis kinetics experiments were reported in April. These experiments covered the measurement of the XeF* (B-X) lifetime and quenching rate constants for the important constituents of XeF laser gas mixtures. Also, lifetime measurements of KrF* (B-X) were reported along with the quenching rate constant for Xe. This report concludes the KrF* quenching, describes measurements of lifetimes and quenching rate constants for the mercury halides (HgBr and HgI), and reports the lifetime of the XeF (C-A) transition.

### 2. KrF* (B-X) Quenching

The experimental system used to measure the quenching rate constants was shown in the previous report. The technique utilized a 2ns pulse of 600 keV electrons from a Febetron 706 to excite an $Ar/F_2$ mixture and produce a short burst of ArF* radiation at 193 nm. This 193 nm pulse entered a quartz cell containing $KrF_2$ vapor and photolyzed the $KrF_2$ producing KrF*(B) + F. The KrF*(B) decay is monitored with a fast photodiode and found to decay with a single exponential lifetime of 6.8 ns. Since the vapor pressure of $KrF_2$ is kept low and since only a small amount of $KrF_2$ is photodissociated, quenching by the dissociation products or the parent molecule can be kept to a negligible level. The rate constants are obtained by using a least squares fit to the

Note: Manuscript submitted April 5, 1979.

measured rate versus quenching gas density data.

Table I presents the measured rate constants for KrF*. All of the gases used for these quenching studies play an important role in either electron-beam-pumped or discharge-pumped KrF lasers. The interpretation of these rate constants are discussed more fully in the accompanying manuscripts published in Applied Physics Letters and the Journal of Applied Physics.

3. HgX* Lifetime and Quenching (X = Br, I)

Because of the recent success in obtaining lasing from the mercury halide molecules, it was deemed important to investigate the kinetics of these molecules. All of the mercury dihalide molecules are stable and commercially available. Reasonable vapor pressures are obtained by raising the dihalide temperature to the 90 - 150$^{o}$C range. To measure the lifetime of these molecules the experiment was modified to the configuration shown in Fig. 1. The photolyzing light pulse is generated in a coaxial diode where the gas mixture is contained within a thin aluminum cylinder and radially excited by the fast Febetron pulse of electrons. Gas mixtures of Ne:Ar:$F_2$ (2000:400:100 torr) were used to produce ArF radiation (193 nm) for photodissociation of $HgBr_2$ and Ar:Kr:$Cl_2$ (2000:200:10 torr) mixtures were used to produce KrCl radiation (222 nm) for photodissociation of $HgI_2$. With both mixtures the pulse width was ~ 2.5 ns FWHM. The pulses were directed into a Suprasil tube filled with a small amount of $HgX_2$ and enclosed in an oven. Photodissociation of the $HgX_2$ produced HgX*(B) + X. The fluorescence from HgX (B-X) decay passed through a narrowband filter and was

monitored by a fast response photomultipler coupled to a digitizer. These signals were plotted on a semilog scale and were found to fit a single exponential decay line. By varying the $HgX_2$ pressure via oven temperature, HgX lifetimes and the quenching of HgX*(B) by $HgX_2$ were determined. These lifetimes and rates are given in Table II. Table II gives values of lifetime of HgCl and quenching of HgCl by $HgCl_2$ which were performed in an analogous manner on another electron beam machine having a somewhat longer pulsewidth. Lifetimes and quenching of HgCl have not yet been done with the fast Febetron system since the wavelength needed to photolyze $HgCl_2$ will not propagate through air.

Quenching rate constants for a number of important laser gas constituents were measured by fixing the $HgX_2$ pressure and adding a wide range of quenching gas pressures. From a least squares fit to the decay rate versus pressure data, two body quenching rate constants have been determined for HgBr* and HgI*. These rate constants are given in Tables III and IV. Of particular concern is the rather large rate constant determined for quenching of HgBr (B) by Hg ($1.3 \times 10^{-10}$ $cm^3$ $sec^{-1}$). Since Hg is a constituent for laser gas mixtures, considerable quenching losses can result. Further discussions of the mercury halide kinetics is contained in the reprint from Applied Physics Letters and in the manuscript submitted to Applied Physics Letters which follow.

4. Lifetime of the C-A Band of XeF

Considerable interest in the C-A transition in XeF has developed recently because of both the possibility that gain might be produced on the transition and the interest that the C level might play in the

kinetics of the B-X laser transition. Fig. 2 shows the potential energy diagram[1] of XeF and the close proximity of the B and C levels. This close proximity raises the possibility of energy transfer from C to B at elevated temperatures. Modeling of such possibilities requires accurate knowledge of the C state lifetime.

To measure the C state lifetime a somewhat different experimental approach was taken than the previous fast photolysis which yielded too few C state molecules. In this case $XeF_2$ vapor was placed in the cylindrical center tube of the coaxial diode and dissociated by electron impact. By using a 460 nm filter ($T_{max}$ = 35% at $\lambda_0$ - 462 nm; $\Delta\lambda$ = 50 nm) to pass only the C-A emission, the fluorescence was detected, displayed on an oscilloscope, recorded, digitized and plotted on a semilog plot from which a decay time was determined. By monitoring the change in decay time as $XeF_2$ pressure was varied from 0.1 - 0.5 Torr, both a measure of the C-A lifetime, 93 $\pm$ 5 ns, and the quenching rate constant for XeF (C) due to $XeF_2$, (1.8 $\pm$ 0.5) $\cdot$ $10^{-10}$ $cm^3$-$sec^{-1}$, were determined. Using this measured lifetime it was possible to correct Kligler's[2] estimated B-C energy defect. The B-C separation is now estimated to be 610 $\pm$ 60 $cm^{-1}$ or .076 eV which agrees with the separation of 0.08 eV measured by Brashears and Setser[3]. Further details can be found in the manuscript to be published in the Journal of Quantum Electronics which follows.

In summary, fast pulse dissociation of the dihalide molecules to nearly instantaneously produce the excited state halide laser molecule provides a clear measurement of excited state lifetimes and collisional quenching rate constants. These measured numbers should allow

realistic modeling of the rare gas and mercury halide lasers.

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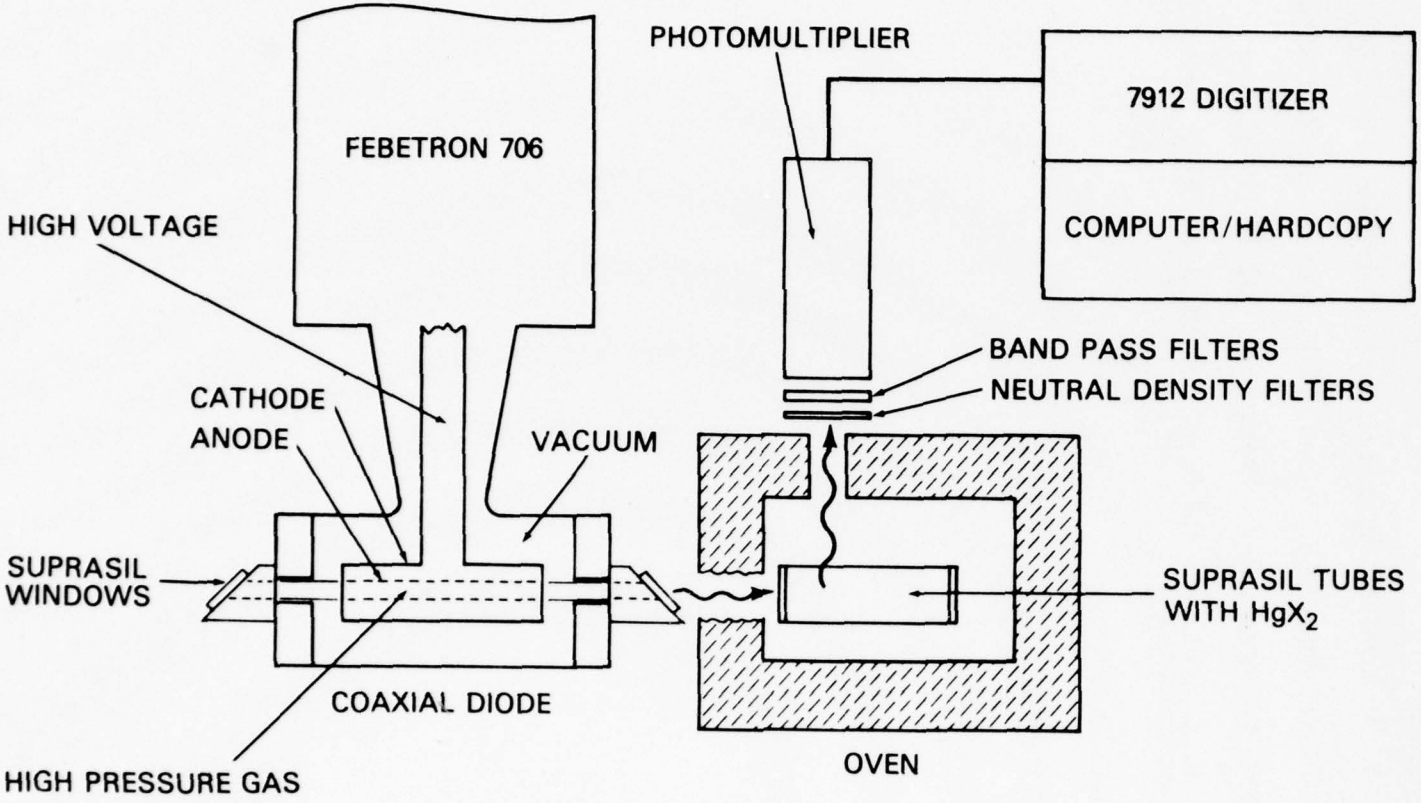



Fig. 1. Apparatus for fast photolysis kinetics experiments.

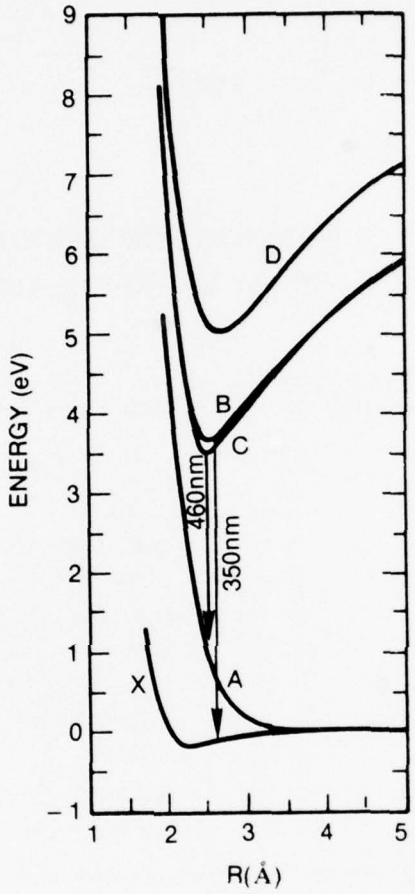



Fig. 2. Potential energy levels of XeF (after Dunning and Hay - ref 1)

TABLE I.

## RATE CONSTANTS FOR QUENCHING OF KrF(B) BY VARIOUS GASES

| Quencher | Rate Constant[a] |
|---|---|
| He | $(3.3 \pm 1.1 \times 10^{-13}$ |
| 2He | $\leq 10^{-33}$ [b] |
| Ne | $(1.6 \pm 0.4) \times 10^{-12}$ |
| 2Ne | $< 10^{-32}$ [b] |
| Ar | $(1.8 \pm 0.6) \times 10^{-12}$ |
| 2Ar | $(1.1 \pm 0.4) \times 10^{-31}$ [b] |
| Kr | $(8.6 \pm 0.7) \times 10^{-12}$ |
| 2Kr | $9.7 \pm 0.8) \times 10^{-31}$ [b] |
| Xe | $> 10^{-9}$ |
| $F_2$ | $(4.8 \pm 0.3) \times 10^{-10}$ |
| $NF_3$ | $(5.2 \pm 0.5) \times 10^{-11}$ |

[a] Units = $cm^3 sec^{-1}$.
[b] Units = $cm^6 sec^{-1}$.

TABLE II.

Radiative lifetimes and di-halide quenching rate constants for the mercury halide (B→X) transitions.

| MOLECULE | $\tau_R$(NS) | $K_Q$($CM^3$–$SEC^{-1}$) |
|---|---|---|
| HgCl(B) | 22.2 ± 1.5 | $9.2 \cdot 10^{-11}$ |
| HgBr(B) | 23.7 ± 1.5 | $1.7 \cdot 10^{-10}$ |
| HgI(B) | 27.3 ± 2.0 | $3.6 \cdot 10^{-10}$ |

TABLE III.

## HgBr(B) QUENCHING RATE CONSTANTS

| QUENCHER | RATE CONSTANT ($CM^3$-$SEC^{-1}$) |
|---|---|
| He | $(5.2 \pm 2.0) \cdot 10^{-13}$ |
| Ne | $(5.3 \pm 1.8) \cdot 10^{-13}$ |
| Ar | $(7.4 \pm 1.5) \cdot 10^{-13}$ |
| Xe | $(3.7 \pm 0.3) \cdot 10^{-12}$ |
| Hg | $(1.3 \pm 0.8) \cdot 10^{-10}$ |
| $N_2$ | $(4.4 \pm 0.5) \cdot 10^{-12}$ |
| $Br_2$ | $(5.5 \pm 0.3) \cdot 10^{-10}$ |
| HBr | $(2.0 \pm 0.4) \cdot 10^{-10}$ |
| $Ch_3Br$ | $(3.9 \pm 0.4) \cdot 10^{-10}$ |
| $CC1_2Br_2$ | $(4.3 \pm 0.4) \cdot 10^{-10}$ |

TABLE IV.

## HgI(B) QUENCHING RATE CONSTANTS

| QUENCHER | RATE CONSTANTS ($CM^3$-$SEC^{-1}$) |
|---|---|
| Ar | $(3.3 \pm 1.5) \cdot 10^{-13}$ |
| Xe | $(7.1 \pm 0.8) \cdot 10^{-13}$ |
| Hg | $< 3 \cdot 10^{-11}$ |
| $N_2$ | $(1.1 \pm 0.2) \cdot 10^{-12}$ |
| HI | $(1.0 \pm 0.2) \cdot 10^{-10}$ |
| $CF_3I$ | $(2.9 \pm 0.3) \cdot 10^{-10}$ |

# New quenching rates applicable to the KrF laser[a)]

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Photolysis of $KrF_2$ has been used to measure the rates of collisional quenching of $KrF(B)$ excimers in two- and three-body collisions with Ar, Kr, and $F_2$. In addition, the $KrF(B \rightarrow X)$ band radiative lifetime was determined to be $6.8 \pm 0.2$ ns. The results are in good agreement with existing theory and demonstrate the importance of quenching to KrF laser performance.

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Recent progress in the development of the KrF uv laser has included improvements in its output power,[1] efficiency,[2] linewidth, and spatial quality of the beam.[3] Since quenching of the laser's upper state imposes limitations on saturation intensity and gas mixture composition, knowledge of fundamental collisional rate constants for the $KrF(B)$ excimer will be required to fully exploit the properties of this molecule as a source of uv stimulated emission.

The rates of quenching of the $KrF(B)$ state by Ar, Kr, and $F_2$ and the radiative lifetime of the $KrF(B \rightarrow X)$ band reported here were determined utilizing pulsed uv photolysis of $KrF_2$ at 193 nm. Absorption of a 193-nm $(ArF[B \rightarrow X])$[1,4] photon by $KrF_2$ results in the immediate collisionless dissociation of the molecule, yielding a $KrF^*(B)$ excimer. Therefore, monitoring the exponential decay of the $B$-state population (in the presence of the desired diluent gas) affords a direct and accurate method of measuring the KrF excimer's rates of quenching by Ar, Kr, and $F_2$.

The experimental apparatus used to photolyze $KrF_2$ is shown schematically in Fig. 1. A 3-ns FWHM beam of 600-keV electrons from a Febetron 706 generator penetrated a 25-$\mu$m-thick Ti foil and irradiated a 99.5% Ar/0.5% $F_2$ (total pressure ≈ 2000 Torr) gas mixture. The fluorescence from this electron-beam-excited plasma, viewed longitudinally through a sapphire window (cutoff ≃ 145 nm) by a Seya-Namioka vacuum spectrograph in first order and Kodak vuv 101-01 film, was found to consist solely of the $ArF(B \rightarrow X)$ band, centered at 193 nm and of ≈ 1.4 nm FWHM. No $F_2^*$ (158 nm)[5] or $Ar_2F^*$ (290 nm)[6] emission was observed. Photodiode studies revealed that the $ArF^*$ emission pulse was characterized by a ~3-ns FWHM and an exponential decay time of 2.5 ns.

After entering the fused silica absorption tube, the 193-nm radiation photolyzed a fraction of the $KrF_2$ molecules in the $KrF_2$-diluent flow, forming $KrF(B)$ excimers. The subsequent radiative decay of the $KrF^*$ population at 249 nm was monitored axially by a narrow-bandpass filter ($T_{max} = 21\%$ at 249 nm; FWHM = 4.5 nm) and a fast-rise-time (~0.5 ns) S-5 biplanar photodiode. Emission waveforms were recorded using a Tektronix 7904 oscilloscope and later digitized by computer for data analysis.

The $KrF_2$ used in these experiments was synthesized according to the technique described by Slivnik and co-workers.[7] Control of the $KrF_2$ vapor pressure was achieved using a dry ice—ethanol bath. Roughly 0.5 Torr $KrF_2$ pressure was maintained in the absorption tube by keeping the bath-cooled $KrF_2$ reservoir at $T \simeq -44$°C. During the experiments, a capacitance manometer recorded the pressure of the $KrF_2$-diluent mixture flowing through the photolytic cell. The rare gases and fluorine used in this work were of research and technical (97%) grades, respectively, and passivation of both the absorption tube and $Ar/F_2$ cell was found to be essential for repeatable results.

Strong photolytic $B \rightarrow X$ fluorescence at 249 nm was obtained with a few tenths of a Torr $KrF_2$ in the system. Signals (1) were observed to decay exponentially over several e-foldings and (2) occurred *only* when both Ar and $F_2$ were present in the excitation cell and $KrF_2$ was *flowing* through the quartz tube. (The rapid dissociation of $KrF_2$ prevented the use of static gas mixtures as was possible with $XeF_2$.[8]) These observations indicate that the photolytic process was indeed the

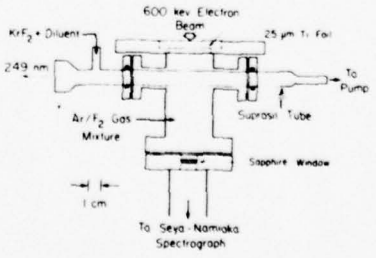



FIG. 1. Schematic diagram of the photolytic apparatus. The Suprasil tube is sealed to the stainless-steel $Ar/F_2$ cell using O-rings and copper gaskets as shown.

[a)]Work supported in part by DARPA.

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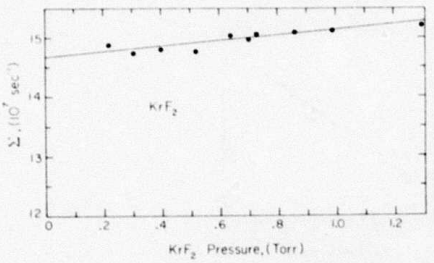



FIG. 2. Graph of decay constant, $\Sigma$, as a function of $KrF_2$ flow pressure. The zero pressure intercept of the least-squares fit to the data (solid line) gives the $KrF(B \rightarrow X)$ radiative lifetime of 6.8 ns.

pumping mechanism for the $KrF(B)$ state and the band-pass filter described earlier ensured that $KrF(B \rightarrow X)$ emission was being monitored.

The analysis and interpretation of the $KrF^*$ waveforms is analogous to that used previously for $XeF^*$.[8,9] Briefly, the decaying $B$-state population is described by a single exponential:

$$[KrF(B)] = [KrF(B)]_{t=0} \exp(-\Sigma t), \tag{1}$$

where

$$\Sigma \equiv \tau_r^{-1} + k_1[Q] + k_2[Q]^2, \tag{2}$$

[ ] indicates particle densities, $\tau_r$ is the $KrF(B \rightarrow X)$ radiative lifetime, and $k_1$ and $k_2$ are the two- and three-body rates of quenching of $KrF(B)$ by the atomic or molecular species $Q$. The single exponential arises from the rapid formation of $KrF(B)$ excimers through the collisionless photodissociation process. By varying the partial pressure of the various diluent gases added to the $KrF_2$ flow, emission waveforms were obtained over a wide range of diluent gas pressures. The falling portion of each waveform was then fit to a single exponential to determine $\Sigma$.

The variation of $\Sigma$ with $KrF_2$ pressure is shown in Fig. 2 where the least-squares fitting of the data to the linear terms of Eq. (2) is represented by the solid line. The zero $KrF_2$ pressure intercept of this line gives the radiative lifetime of the $KrF(B \rightarrow X)$ band as $6.8 \pm 0.2$ ns where the standard deviation and experimental uncertainty of the data in Fig. 2 are represented by the 0.2-ns error. Although this value is lower than that obtained recently using an ArF laser to photodissociate $KrF_2$,[10] the $\approx 20\%$ difference between the two results is not understood and uncertainties due to the detection system have been eliminated. The present value is in excellent agreement with the theoretical lifetime of 7.0 ns calculated by Hay and Dunning.[11,12] Two-body destruction of $KrF^*$ by $KrF_2$ was found to proceed at the rate of $(1.4 \pm 0.4) \times 10^{-10}$ cm$^3$ s$^{-1}$.

Figure 3 shows the dependence of $\Sigma$ on diluent pressure for $KrF(B)$ quenching by Kr where the quadratic component, representing three-body quenching of the excimer by 2 Kr, becomes clearly evident. The two- and three-body quenching rates for Kr obtained from Fig. 3 are $k_1 = (8.6 \pm 0.7) \times 10^{-12}$ cm$^3$ s$^{-1}$ and $k_2 = (9.7 \pm 0.8) \times 10^{-31}$ cm$^6$ s$^{-1}$, respectively, where again the uncertainty in the rate constants shown in parentheses reflects the standard deviation of the data coupled with an estimated experimental error of $\lesssim 10\%$ per data point. By analyzing e-beam-pumped $Ar/Kr/F_2$ mixtures, Rokni, Jacob, and Mangano[13] inferred a two-body rate almost an order of magnitude smaller ($k_1 < 1.6 \times 10^{-12}$ cm$^3$ s$^{-1}$ for $\tau_r = 6.9$ ns) than that reported here. In contrast, the rate given above for three-body quenching of $KrF(B)$ by krypton is in good agreement with the theoretical prediction of Shui[14] ($5 \times 10^{-31}$ cm$^6$ s$^{-1}$) and the result of Rokni *et al.* ($6.4 \times 10^{-31}$ cm$^6$ s$^{-1}$).

The absence of a curve crossing between the $X$ (or $A$) ground state and $B$ excited level of KrF [12] suggests that the Kr two-body quenching rate is due to mixing collisions between the $B$ and $C$ states:

$$KrF(B) + Kr \rightleftarrows KrF(C) + Kr. \tag{3}$$

Such a mechanism is consistent with the proximity of the two states in energy[11] and may partially account for the discrepancy in reported Kr two-body rates. For the gas mixtures used in Ref. 13, the KrF $C$ and $B$ states are likely to be equilibrated due to thermal collisions [Eq. (3)]. Thus, a smaller *net* quenching rate than that presented above may be expected since the $C$ and $B$ levels act essentially as one state. However, the low Kr pressures ($\leq 200$ Torr) and short pulse lengths employed here make equilibrium unlikely, favoring a larger loss rate for $KrF(B)$.

The physical interpretation of three-body quenching of $KrF(B)$ by Kr has been convincingly shown to be the process of $Kr_2F^*$ trimer formation[13]:

$$KrF(B) + 2\,Kr \rightarrow Kr_2F(^2B_2) + Kr \tag{4}$$

and culminates in the emission of a $Kr_2F(^2B_2 \rightarrow A_1)$ photon at ~415 nm. Shui's calculations invoke this mechanism and the good agreement between theory and experiment seems to further support this conclusion.

Similar data for Ar and $F_2$ are shown in Figs. 4 and 5. From these curves, the rate constants were determined to be $k_1(Ar) = (1.8 \pm 0.6) \times 10^{-12}$ cm$^3$ s$^{-1}$, $k_2(Ar) = (1.1 \pm 0.4) \times 10^{-31}$ cm$^6$ s$^{-1}$, and $k_1(F_2) = (4.8 \pm 0.3) \times 10^{-10}$ cm$^3$ s$^{-1}$. Again, the argon three-body rate is in excellent agreement with theory ($0.9 \times 10^{-31}$ cm$^6$ s$^{-1}$)[14] suggesting the formation of stable $ArKrF^*$ trimers. To our knowledge, the two-body quenching rate for Ar has

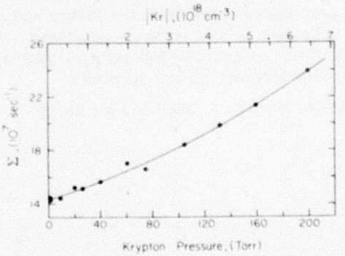



FIG. 3. Dependence of $\Sigma$ on Kr pressure where, again, the solid line represents the least-squares fit of the data to Eq. (2).

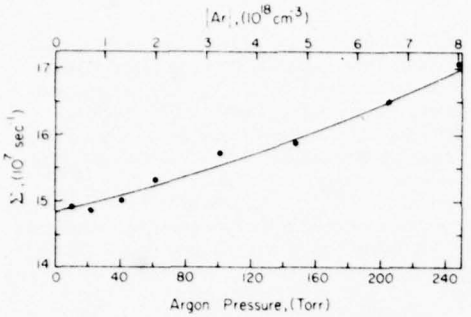



FIG. 4. Variation of Σ with [Ar]. Experimental difficulties in mixing $KrF_2$ with Ar prevented the study of higher Ar pressures.

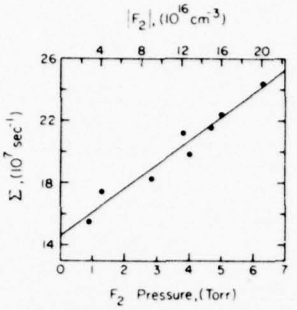



FIG. 5. Plot of decay constant versus $F_2$ pressure. The slope of the solid line yields the rate of two-body quenching of KrF* by $F_2$.

not been previously measured and yet is one of the main sources of KrF* quenching in electron-beam-pumped $Ar/Kr/F_2$ gas mixtures. For example, for a typical mixture composition of 95% Ar/4.8% Kr/0.2% $F_2$ (total pressure ≈ 2100 Torr), collisional deactivation of KrF($B$) by two-body collisions with Ar accounts for 17% of the total quenching loss, three-body Ar collisions are responsible for 66%, $F_2$ for 11%, and all Kr collisions for 6%. Thus, collisional quenching of KrF excimers by Ar greatly exceeds that of the other gas constituents and it is quite possible that replacing argon by neon will result in a significant improvement in laser performance as it has for the XeF laser.[8,15] From the measured quenching rates and letting $\sigma_{SE}\tau = 1.75\times10^{-24}$ cm$^2$s$^{-1}$ (Ref. 16), the saturation intensity of the KrF laser, given the gas mixture mentioned above, is 2.6 MW cm$^{-2}$. In the absence of stimulated emission, a KrF($B$) excimer has an effective lifetime of 1.5 ns.

Finally, preliminary measurements of the quenching rate of KrF($B$) by He give a two-body rate of $\approx 3\times10^{-13}$ cm$^3$s$^{-1}$, indicating that the rate constants reported in this paper are not products of vibrational quenching of the $B$-state manifold. That is, ArF* photolytic dissociation of $KrF_2$ appears to produce only low vibrational levels of the KrF($B$) state.

In summary, the KrF($B\rightarrow X$) radiative lifetime and the two- and three-body collisional quenching rates of KrF($B$) by Kr, Ar, and $F_2$ have been measured by photolyzing $KrF_2$ in the presence of the desired diluent and monitoring the resulting $B\rightarrow X$ radiation at 249 nm. The results are in good agreement with theory and demonstrate the important role of excimer quenching in determining the laser's saturation intensity and power loading of the mirrors forming the optical cavity.

The authors wish to thank L. J. Palumbo for aid in digitizing the data and C. Mullins and D. Epp for technical assistance.

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# KrF(*B*) quenching by He, Ne, Xe, and $NF_3$[a)]

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Flash photolysis of $KrF_2$ at 193 nm has been utilized to directly measure the quenching rate constants of KrF(*B*) molecules by He, Ne, Xe, and $NF_3$. Investigation of the large rate constant measured for the KrF(*B*) + Xe reaction ($\gtrsim 10^{-9}$ $cm^3 sec^{-1}$) verified the XeF(*B*) state to be a product.

PACS numbers: 31.50.+w, 32.50.+d, 82.50.Et, 82.20.Pm

## I. INTRODUCTION

Experiments of the past year have confirmed the importance of exciplex quenching on XeF and KrF laser performance.[1-3] Collisional losses due to formation of rare gas–fluoride trimers ($R_2F^*$, $R$ = Xe, Kr) or $R$F(*C*) state molecules have been proposed to account for the reported quenching rates which, in turn, are responsible for laser mixture restrictions and require large circulating powers in the optical cavity. For example, argon was commonly used as the diluent in e-beam-pumped XeF laser mixtures until the discovery of the superior quenching[3] and optical absorption[4] characteristics of neon. The possibility of a similar situation existing for the KrF laser prompted the experimental work described below.

For KrF*, the rates of quenching of KrF(*B*) molecules by Ar, Kr, and $F_2$[5,9,10] and the radiative lifetime of the *B* state[10,11] have been measured previously. This paper describes the direct measurement of the deactivation rates of KrF(*B*) molecules by He, Ne, Xe, and $NF_3$, using pulsed uv photolysis of $KrF_2$. Two obvious advantages of this experimental technique are (1) quenching of the KrF exciplex population by electron impact can be ignored[12] and (2) the various quenching atoms or molecules of interest may be studied independently of one another.

Most of the rates reported here have been measured for the first time. Additional observations suggest that quenching of KrF(*B*) by Xe results in the formation of XeF(*D*) molecules followed by collisional or radiative transfer to the *B* state of XeF.

Section II discusses the experimental technique and data analysis used in this investigation, while Sec. III presents the results for the quenchers mentioned above. Deactivation of KrF(*B*) by Xe and KrF(*B*) → XeF* excitation transfer are discussed in Sec. IV and, finally, Sec. V reviews the conclusions of this study.

## II. EXPERIMENTAL TECHNIQUE AND DATA ANALYSIS

A schematic diagram of the experimental apparatus used to photolyze $KrF_2$[10] is shown in Fig. 1. Briefly, ArF(*B*→*X*) emission from electron-beam-irradiated $Ar/F_2$ gas mixtures (99.5% Ar/0.5% $F_2$; $p_{total} \simeq 2000$ Torr) photolyzed $KrF_2$ molecules in a $KrF_2$-diluent stream flowing through a Suprasil (VUV quality) quartz tube. A Seya-Namioka vacuum spectrograph (in first order) recorded the ArF* emission spectrum shown in Fig. 2 by viewing the fluorescence through a sapphire window ($\lambda_{cutoff} \sim 145$ nm). The spectral FWHM shown in Fig. 2 is $\sim 1.2$ nm which corresponds to $\Delta E \sim 320$ $cm^{-1}$. Using a solar blind photodiode, the ArF* emission pulse was found to closely follow the electron-beam current.

As mentioned in Ref. 10, the $KrF_2$ used in these experiments was synthesized according to the technique described by Slivnik *et al.*[13] The $KrF_2$ crystals were contained in a Pyrex vessel which in turn was immersed in a dry-ice–alcohol bath. Mixing of the desired quenching gas with the $KrF_2$ vapor was accomplished by flowing the gas over the crystals as shown in Fig. 1. The $KrF_2$ vapor pressure was maintained at $\simeq 0.5$ Torr in these experiments by keeping the dry-ice–alcohol bath at $T \sim -44$ °C. Diluent was then added to raise the total flow pressure (measured by a capacitance manometer) to the desired value. Although improvements have been made in the flow system,[10] gas mixing difficulties prevented study of diluent pressures in excess of 1000 Torr. All gas-handling lines were 6-mm o.d. Teflon tubing. Also, flowing $KrF_2$ vapor through the quartz tube for several minutes prior to the experiments was found to be important for obtaining repeatable results. Details of the quartz absorption tube and $Ar/F_2$ photolytic cell have been published in a recent paper.[10]

Subsequent to absorbing a 193-nm photon, a $KrF_2$ molecule dissociates,[10,14] creating a KrF(*B*) exciplex by the collisionless process:

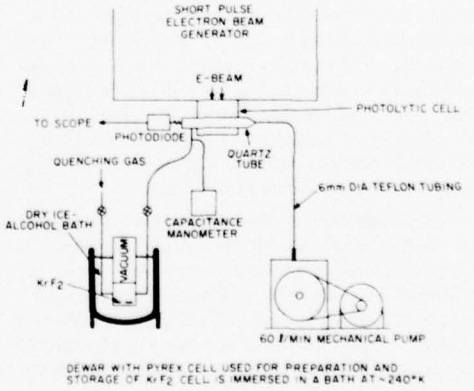



FIG. 1. Partial schematic diagram of the experimental apparatus used to photolyze $KrF_2$ and measure rates of quenching of KrF(*B*) by various gases. Details of the construction of the quartz absorption tube and $Ar/F_2$ photolytic cell are given in Ref. 10.

[a)]Work supported in part by DARPA.

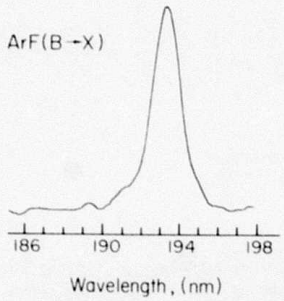



FIG. 2. ArF($B \rightarrow X$) emission spectrum at 193 nm from e-beam-excited $Ar/F_2$ gas mixtures. The detection system resolution was ~0.3 nm and the vertical scale is logarithmic.

$$KrF_2 + h\nu(6.4\text{ eV}) \rightarrow KrF_2^* \rightarrow KrF(B) + F + \Delta E\,(0.4\text{ eV}). \quad (1)$$

The KrF($B \rightarrow X$) 249-nm emission waveforms were monitored by a S-5 surface photodiode (~0.5 nsec rise time) and narrow bandpass filter ($T_{max} = 21\%$ at 249 nm; $\Delta\lambda = 4.5$ nm) and the traces were recorded on film using a Tektronix 7904 oscilloscope. The measured rapid rise in KrF($B \rightarrow X$) flourescence following the onset of ArF excitation suggested that the dissociation of $KrF_2^*$ by Eq. (1) was rapid (< 1 nsec).

Since no collisions are required to *form* the KrF($B$) population, the extraction of quenching rates from the pressure-dependent fluorescence data is straightforward. Simple kinetic rate equations[3,4,10] predict-decaying waveforms described by a single exponential decay constant $\Sigma$:

$$\Sigma \equiv \tau_r^{-1} + k_1[Q] + k_2[Q]^2 + k_3[KrF_2], \quad (2)$$

where $\tau_r$ is the KrF($B \rightarrow X$) radiative lifetime of 6.8 nsec,[10] $k_1$ and $k_2$ are the two- and three-body rates of quenching of KrF($B$) by the diluent $Q$, and $k_3$ is the rate constant for quenching of KrF($B$) by $KrF_2$ ($1.4 \times 10^{-10}$ cm$^3$ sec$^{-1}$).[10] Experiments confirm the single exponential nature of the florescence signals as shown in Fig. 3 which presents typical ArF* and KrF($B \rightarrow X$) emission waveforms. The ArF trace is characterized by a 2.5-ns decay constant, whereas the KrF signal, corresponding to a 0.5 Torr $KrF_2$ plus 21 Torr Ne gas mixture, has a decay constant of ≲6.8 nsec.

To determine the KrF($B$) quenching rate for a given diluent, the fluorescence waveforms for various diluent pressures were digitized from the oscillograms and plotted on semilog scales by a CDC 3800 computer. Then, for each pressure, $\Sigma$ was determined by fitting the falling portion of the signal to a single exponential and correcting for the nonzero decay time of the ArF* excitation pulse by deconvolving the pump and KrF fluorescence waveforms. This correction is < 20% for signal decay times ≥5 nsec. From a graph of $\Sigma$ versus $[Q]$, determinations of $k_1$ and $k_2$ could then be made.

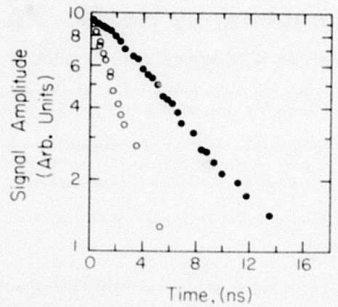



FIG. 3. ArF($B \rightarrow X$) (○) and KrF($B \rightarrow X$) (●) fluorescence waveforms typical of those encountered in these experiments. The ArF* emission was observed from e-beam-excited 2000 Torr Ar/10 Torr $F_2$ mixtures with a bandpass filter ($T_{max} = 20\%$ at 198.0 nm; $\Delta\lambda = 20$ nm) and a solar blind photodiode. The KrF radiation was due to a 0.5 Torr $KrF_2$–21 Torr Ne gas mixture flowing through the reaction cell. $T = 0$ identifies the time of peak e-beam current.

## III. RESULTS AND DISCUSSION

The dependence of $\Sigma$ on helium pressure is shown in Fig. 4 where the least-squares fitting of the data points is represented by the solid line. The slope of this line gives the rate of quenching of KrF($B$) by a two-body collision with He as $(3.3 \pm 1.1) \times 10^{-13}$ cm$^3$ sec$^{-1}$ where the experimental and statistical uncertainty of the rate constant is given in the parentheses. Due to the ~15% per data point uncertainty in $\Sigma$, the three-body quenching rate for He was estimated to be ≲ $10^{-33}$ cm$^6$ sec$^{-1}$. Given 2 atm of He in a discharge-pumped KrF laser mixture, the rate of helium quenching of KrF($B$) is $1.6 \times 10^7$ sec$^{-1}$, thus reducing the effective lifetime of a KrF exciplex by 10% to 6.1 nsec.

The contrast between Ne and Ar as KrF quenchers is clearly shown in Fig. 5 where the Ar data was taken from Ref. 10. Again, the solid lines represent the least-squares fitting of the data points. For neon, the two-body KrF($B$) quenching rate, $k_1(Ne) = (1.6 \pm 0.4) \times 10^{-12}$ cm$^3$ sec$^{-1}$, is simi-

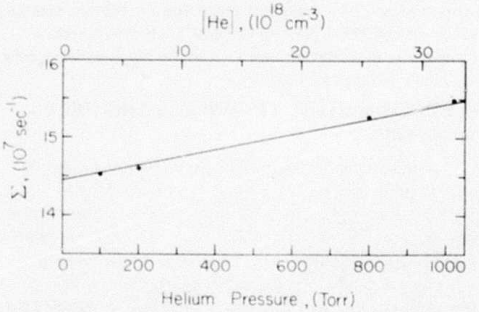



FIG. 4. Graph of $\Sigma$ versus He pressure. The solid line represents the least-squares fit of the data points to the linear terms of Eq. (2). The experimental uncertainty in $\Sigma$ per point is ≲15% and the error in $p_{He}$ is negligible.

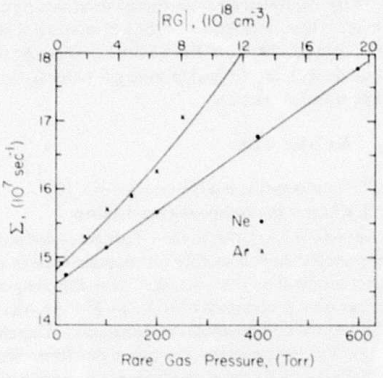



FIG. 5. Comparison of quenching of KrF($B$) by Ne and Ar. The different vertical axis intercepts are due to slightly different $KrF_2$ flow pressure used in the two experiments.

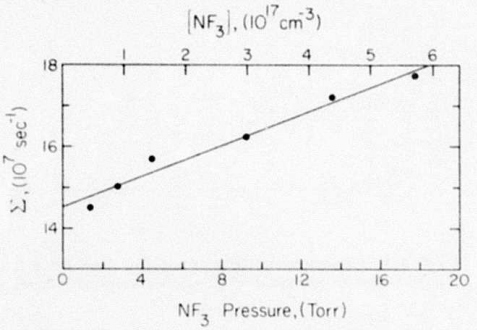



FIG. 6. Variation of decay constant $\Sigma$ with $NF_3$ concentration. The slope of the solid line is $5.2\times10^{-11}$ cm$^3$ sec$^{-1}$ and again the error in the decay constant per data point is $\sim 15\%$.

lar to that measured for Ar, $(1.8\pm0.6)\times10^{-12}$ cm$^3$ sec$^{-1}$. The key difference between the two diluents lies in the three-body rate constants which for neon is estimated to be $<10^{-32}$ cm$^6$ sec$^{-1}$ while for argon, $k_3=(1.1\pm0.4)\times10^{-31}$ cm$^6$ sec$^{-1}$.

To compare argon and neon as diluents for e-beam-pumped KrF lasers, consider a typical 95% Ar/4.8% Kr/0.2% $F_2$ gas mixture with a total pressure of 2100 Torr. From the KrF* quenching rates for Ar, Kr, and $F_2$ determined previously,[10] Ar accounts for 83% of the collisional losses, Kr for 6%, and $F_2$ for 11%. Substituting neon for argon (and doubling the diluent concentration to maintain constant electron energy deposition) reveals that neon accounts for 86% of the $B$-state quenching losses. Therefore, exchanging argon for neon does not substantially change diluent quenching and so any advantage of neon over argon (as is the case for XeF[5,15]) will depend on KrF formation kinetics in Ne/Kr/$F_2$ mixtures.

Figure 6 shows the results for $NF_3$ where the two-body quenching rate is $(5.2\pm0.5)\times10^{-11}$ cm$^3$ sec$^{-1}$. As with XeF*, $NF_3$ quenches KrF($B$) molecules roughly an order of magnitude less effectively than does $F_2$. However, it is well known that $NF_3$ does not perform as well as $F_2$ in KrF laser systems, which is probably due to $NF_3$ (or its fragments, e.g., $NF_2$) absorption at 250 nm[14] or to the importance of $F_2^*$ molecules in the exciplex formation chain.

## IV. KrF($X$) QUENCHING BY Xe AND KrF($B$)→XeF* EXCITATION TRANSFER

Xenon was found to quench KrF($B$) molecules at a surprisingly large rate as shown by Fig. 7. The slight spread of the data points does not accurately reflect the experimental error due to the low xenon pressures ($\leq$ 1.5 Torr) and large decay constants involved. For this reason, error bars are given in Fig. 7. Each data point has been corrected for uncertainties in the $KrF_2$ pressure which arose from slight variations in the temperature of the alcohol–dry-ice bath used to control the $KrF_2$ flow pressure. Fluctuations in the bath temperature of as little as 1 °C caused $KrF_2$ pressure deviations of 0.05 Torr. This, in turn, created an uncertainty of $\pm$0.05 Torr in the xenon pressure since the capacitance manometer monitored the total flow pressure ($p_{KrF_2}+p_{Xe}$).

Including the experimental uncertainties mentioned above and statistical errors arising from the least-squares fit, the data of Fig. 7 give a lower limit for the rate constant for two-body quenching of KrF($B$) by Xe of $\sim10^{-9}$ cm$^3$ sec$^{-1}$. Preliminary studies by other investigators[16] have yielded a half-quenching pressure of $\sim$1.8 Torr or $k=2.5\times10^{-9}$ cm$^3$ sec$^{-1}$ (setting $\tau_{KrF(B)}=6.8$ nsec) for this reaction. Our result is also consistent with the findings of Rokni and coworkers[9] who found the analogous displacement reaction: $ArF^*+Kr\rightarrow KrF^*+Ar$ to proceed with a rate constant of $1.6\times10^{-9}$ cm$^3$ sec$^{-1}$.

In an attempt to determine the products of the KrF($B$) + Xe quenching mechanism, fluorescence at 350 nm

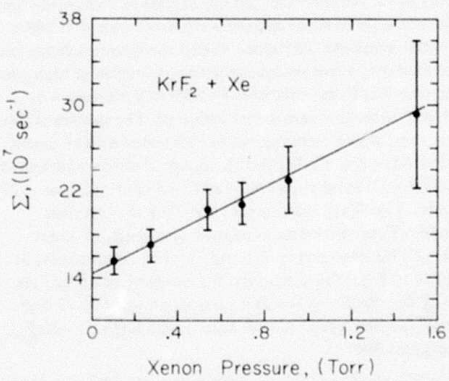



FIG. 7. Quenching data for xenon including corrections for the 2.5-nsec decay time of the ArF* fluorescence. The error bars indicate the estimated experimental errors for the low pressure measurement. The solid line denotes the linear least-squares fitting of the data points. The lower limit for $k_1$(Xe) is $10^{-9}$ cm$^3$ sec$^{-1}$.

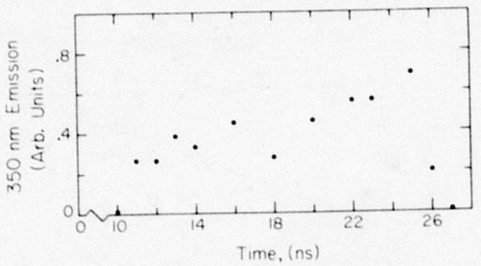



FIG. 8. Time dependence of XeF($B \rightarrow X$) emission observed from 0.5 Torr $KrF_2$–14 Torr Xe mixtures. $T=0$ corresponds to peak ArF* fluorescence.

[emitted by low-lying XeF($B$) vibrational levels] from $KrF_2$-Xe photolyzed mixtures was monitored by a narrow bandpass filter ($\lambda_0 = 348$ nm, $\Delta\lambda = 9.4$ nm) and a S-5 photodiode. XeF($B \rightarrow X$) emission was observed but, as shown in Fig. 8, the onset of the fluorescence was delayed by $\sim 10$ nsec from the peak of the ArF pulse. This effect may be explained by considering the relative energies of the KrF($B$), XeF($D$), and XeF($B$) states. Using the spectroscopic constants given by Tellinghuisen *et al.*[17] for the $D$ level of XeF and those for KrF($B$) calculated by Hay and Dunning,[18] several close energy coincidences ($< 300$ cm$^{-1}$) between the KrF($B$, $v'=0,1$) and XeF($D$, $v''=12$–14) vibrational levels were found. Thus, those near-resonant states allow for KrF($B$)$\rightarrow$XeF($D$) excitation transfer followed by collisional or radiative cascading to the XeF($B$) state. Since the mixture pressure of Fig. 8 is low ($< 15$ Torr), collisional relaxation of the XeF($B$ and $D$) state vibrational manifolds will be slow, in agreement with the observed time delay. Also, the 10-nsec lapse before the $B \rightarrow X$ radiation was detected minimizes the possibility of direct KrF($B$)$\rightarrow$XeF($B$, low $v''$) transfer. Second, the 350-nm radiation is emitted over $\sim 16$ nsec, indicating a relatively slow "feeding" of low-lying XeF($B$) vibrational levels.

Although it is desirable to directly observe the XeF($D \rightarrow X$) emission at 260 nm, a filter of sufficiently narrow response to eliminate the KrF($B \rightarrow X$) band at 249 nm was not available. Therefore, to test the kinetic scheme proposed above, experiments involving excitation of high pressure $Ar/Kr/F_2$ gas mixtures by a 450-kV 50-nsec $\sim 1$-kA cm$^{-2}$ electron beam were conducted. The spectra of these mixtures, which were typical of those used as KrF lasers (95% Ar/4.8% Kr/0.2% $F_2$), consisted of the well-known KrF($B \rightarrow X$) band at 249 nm and $C \rightarrow A$ continuum at $\sim 275$ nm [cf. Fig. 9(a)]. Adding just one Torr of Xe to this $Ar/Kr/F_2$ mixture produced intense XeF($B \rightarrow X$) and ($D \rightarrow X$) fluorescence at 351 and 260 nm, respectively, as shown in Fig. 9(b). Certainly, the intensity of neither the $B \rightarrow X$ nor the $D \rightarrow X$ bands is characteristic of Ar/1 Torr $Xe/F_2$ gas mixtures but is consistent with KrF($B$)$\rightarrow$XeF($D$) energy transfer.

The large ratios of [Ar]/[Xe] and [Kr]/[Xe] were chosen so that the kinetic chain forming KrF($B$) would be negligibly perturbed by the addition of 1 Torr Xe to the gas mixture of Fig. 9(a). In electron-beam excited rare gas plasmas, the bulk of the deposited electron energy produces rare gas atomic ions (in this case, $Ar^+$). For the gas mixture of Fig. 9(a) (2000 Torr Ar/100 Torr Kr/5 Torr $F_2$), these $Ar^+$ ions rapidly collide with the Ar background gas to form $Ar_2^+$. The charge transfer reaction

$$Ar_2^+ + Kr \xrightarrow{k} Kr^+ + 2Ar \tag{3}$$

is known[19,20] to proceed at a large rate ($k \simeq 6 \times 10^{-10}$ cm$^{-3}$ sec$^{-1}$), whereas the competing mechanism ($Ar_2^+ + Xe \rightarrow Xe^+ + 2Ar$), due to the 1 Torr Xe added in Fig. 9(b), is expected to have a smaller rate constant due to the $\sim 1$-eV exothermicity of the reaction.[19] Also, the competing ion recombination processes $Kr^+$(or $Xe^+$) $+ F^- + Ar \rightarrow KrF^*$-(or XeF*) + Ar have comparable rate constants.[21] Therefore, the spectra of Fig. 9 support the conclusion that formation of both the XeF($D$ and $B$) states result from the quenching of KrF($B$) by Xe.

A summary of the experimentally determined rate constants for quenching of the KrF($B$) state by various gases is given in Table I. Some of the rate constants determined by other investigators depend on $\tau_r$ and so the values presented in Table I assume $\tau_r = 6.8$ nsec.

Since the experimental technique used in the present work measures *only* the rate at which the KrF($B$) state molecules are destroyed, the products of the various quenching collisions shown in Table I are not known. In a previous paper,[10] the discrepancy between the reported two-body rate constants for Kr was attributed to mixing between the KrF $B$ and $C$ states:

$$KrF(B) + Q \rightleftarrows KrF(C) + Q, \tag{4}$$

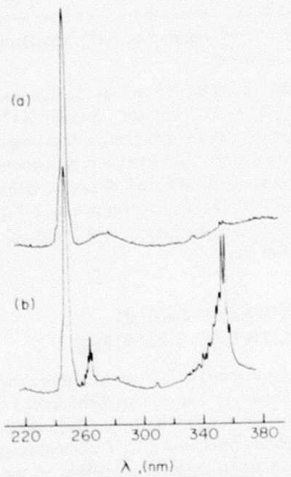



FIG. 9. Spectra recorded from e-beam-excited mixtures: (a) 2000 Torr Ar/100 Torr Kr/5 Torr $F_2$ mixture, clearly showing the KrF 249- and 275-nm bands and (b) same mixture as (a) with a Torr of Xe added. Vertical scale for both spectral profiles is logarithmic.

TABLE I. Rate constants for quenching of KrF($B$) by various gases.

| Quencher | Rate constant[a] | Reference |
|---|---|---|
| He | $(3.3\pm1.1)\times10^{-13}$ | This work |
| 2He | $\leq10^{-13b}$ | This work |
| Ne | $(1.6\pm0.4)\times10^{-12}$ | This work |
| 2Ne | $<10^{-12b}$ | This work |
| Ar | $(1.8\pm0.6)\times10^{-12}$ | 10 |
| 2Ar | $(7.6\pm0.7)\times10^{-12b}$ | 7, 9 |
| | $(1.1\pm0.4)\times10^{-11}$ | 10 |
| Kr | $<1.6\times10^{-12}$ | 9 |
| | $(8.6\pm0.7)\times10^{-12}$ | 10 |
| Ar+Kr | $6.2\times10^{-13b}$ | 9 |
| 2Kr | $6.5\times10^{-13b}$ | 9 |
| | $2.9\times10^{-11}$ | 5 |
| | $(9.7\pm0.8)\times10^{-11}$ | 10 |
| Xe | $>10^{-9}$ | This work |
| $F_2$ | $7.4\times10^{-10}$ | 9 |
| | $5.7\times10^{-10}$ | 5 |
| | $(4.8\pm0.3)\times10^{-10}$ | 10 |
| $NF_3$ | $(5.2\pm0.5)\times10^{-11}$ | This work |

[a]Units = $cm^3$ $sec^{-1}$.

[b]Units = $cm^6$ $sec^{-1}$.

where, again, $Q$ is the diluent gas of interest. This process may also be important for the quenching reactions studied here and particularly for He and Ne where large diluent concentrations were used to measure the desired rate constants.

## V. CONCLUSIONS

The rates of quenching of KrF($B$) by He, Ne, Xe, and $NF_3$ have been reported in this paper. Simple exponentially decaying waveforms were obtained by photolyzing $KrF_2$ in the presence of the desired diluent with 193-nm radiation. The results underscore the significance of quenching studies to an understanding of basic decay processes of the exciplex population.


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# $HgX(B)$ radiative lifetime by fast photolysis of $HgX_2$ ($X = Br,I$)[a]

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The radiative lifetimes of the $HgBr(B)$ and $HgI(B)$ states have been determined to be $23.7 \pm 1.5$ and $27.3 \pm 2.0$ ns, respectively, by fast photolysis of $HgBr_2$ and $HgI_2$ using e-beam-excited ArF and KrCl fluorescence. This experimental approach generated simple exponential decay curves which facilitated measurement of the lifetimes and the rate of quenching of the excited $HgX(B)$ states by $HgX_2$ ($X = Br$ or I) molecules. Although the lifetimes reported here are in agreement with previous experimental measurements, they contradict a recently published theoretical calculation.

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Laser emission on the $HgBr(B \rightarrow X)$ transition has been produced by electron-beam pumping of mixtures of Ar, Xe, Hg, and HBr[1]; by electron beam sustained discharge pumping of Ar, Hg, and $BrCCl_3$ mixtures[2]; by using an ArF laser (193 nm) to photodissociate $HgBr_2$ to produce $HgBr(B)$ [3]; and by dissociating $HgBr_2$ in a fast transverse electric discharge.[4,5] These latter experiments also were able to dissociate $HgI_2$ to produce lasing on the $HgI(B \rightarrow X)$ band.

The technique of producing an $HgX$ ($X = Br, I$) laser by dissociating the $HgX_2$ molecule is of practical interest since the $HgX$ molecule in its ground state may recombine with a free halogen atom to form $HgX_2$, and, therefore, the laser can operate repetitively in a sealed system. Energies of ~7 mJ already have been obtained from these rare-gas/$HgX_2$ salt vapor discharges,[5] but optimal design requires knowledge of excited-state lifetimes and quenching rates.

In recent experiments, Eden[6] has measured the radiative lifetime of the $HgCl(B \rightarrow X)$ transition as $22.2 \pm 1.5$ ns and the quenching rate of $HgCl(B)$ by $HgCl_2$ as $9 \times 10^{-11}$ $cm^3$ $s^{-1}$. In this paper, the lifetimes of $HgX(B)$ ($X = Br, I$) and the rate constants for quenching of $HgX(B)$ by $HgX_2$ have been determined by rapidly photodissociating $HgX_2$. Spontaneous emission from either the ArF or KrCl molecules was used as the photolytic pump source. The results reveal excited-state lifetimes which do not conform to the theoretical calculations of Duzy and Hyman.[7]

The experimental system used to make the lifetime and quenching measurements is shown in Fig. 1. A Febetron 706 is used to deliver a 3-ns 600-kV voltage pulse to an impedance-matched diode of coaxial geometry similar to that reported by Bradley and co-workers.[8] The 25-$\mu$m-thick aluminum cylindrical anode was filled either with $Ne:Ar:F_2$ (2000 : 400 : 10 Torr) to produce ArF radiation ($\lambda = 193$ nm)[9] for photodissociation of $HgBr_2$ or with $Ar:Kr:Cl_2$ (2000 : 200 : 10 Torr) gas mixtures to produce KrCl radiation[10] ($\lambda = 222$ nm) for photodissociation of $HgI_2$. Although e-beam excitation of $Ar/Kr/Cl_2$ mixtures also produces $ArCl^*$ (175 nm) and $Cl_2(E \rightarrow B)$ radiation at 258 nm in addition to KrCl fluorescence, the $ArCl^*$ emission is absorbed in air before reaching the heated cell and $HgI_2$ is transparent to the $Cl_2$ 258-nm radiation.[11] The electron-beam cell was thoroughly passivated with technical purity $F_2$ or $Cl_2$, and research-grade rare gases were used for the high-pressure mixtures. Suprasil quartz windows were mounted at each end of the e-beam cell. The rare-gas—halide excimer radiation was monitored at one end of the cell by a fast-rise-time (~0.5 ns) ITT F4018 photodiode with a S-5 surface and a narrow-bandpass filter ($\lambda_0 = 198$ or 221 nm; $\Delta\lambda \sim 20$ nm). These filters transmit 19% at 193 nm and 25% at 222 nm, respectively. Both the ArF (193 nm) and KrCl (222 nm) spontaneous emission waveforms were observed to have a FWHM of ~2.5 ns and decay times of ~3—4 ns.

Radiation from the other end of the diode was directed into an oven-heated quartz cell containing a few mg of the desired mercuric halide salt. These optical cells were evacuated to $< 10^{-5}$ Torr and baked prior to filling with $HgX_2$. Also, the $HgX_2$ was distilled over a hydrogen flame to drive off impurities before being sealed. In some cells, 100 Torr of He also was added to study the effects of vibrational relaxation on the apparent $HgX^*$ radiative lifetime. Oven temperature was monitored carefully by a thermometer and the $HgX_2$ particle density was determined from vapor-pressure curves.[12]

The fluorescent emission from the $HgX(B)$ molecules

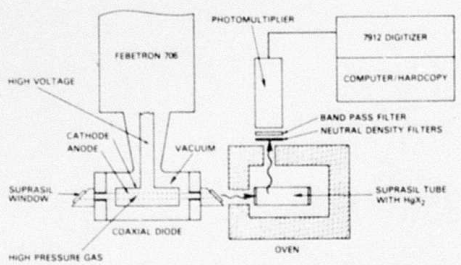



FIG. 1. Partial schematic diagram of the experiment. Details of the experimental technique are given in the text.

[a]Work supported in part by DARPA.

was monitored at right angles to the cell by an EMI 9818QB photomultiplier mounted in both a rf- and x-ray-shielded housing. The spectral regions of interest (500 nm for HgI and 440 nm for HgBr) were selected by bandpass filters transmitting over about 15 nm bandwidth and nearly centered about the wavelength of interest. It was necessary to attenuate the $HgX(B \rightarrow X)$ fluorescence signals with neutral density filters to avoid saturating the photomultiplier. The time response of the photomultiplier system was calibrated by allowing it to view the excitation pulse from the diode through the ultraviolet filters described earlier. The HgBr and HgI $B \rightarrow X$ spontaneous emission signals from the photomultiplier were recorded using a Tektronix 7912 digitizer, Tektronix 1152 computer, and a hard copy printer. The curves were then redigitized and plotted on semilog scales using a CDC 3800 computer.

Maya[11] has measured the uv absorption spectra of both $HgBr_2$ and $HgI_2$ and found that the threshold wavelengths for photoproduction of the HgBr($B$) and HgI($B$) states are ~205 and 235 nm, respectively. For example,

$$HgBr_2 + h\nu \ (\lambda \lesssim 208 \text{ nm}) \rightarrow HgBr_2^* \rightarrow HgBr(B) + Br. \tag{1}$$

Therefore, since the process pumping the $HgX(B)$ state is collisionless and the pulse widths of the ArF and KrCl pump radiation are considerably less than the estimated $B$-state lifetimes,[7] then to a first approximation the temporal dependence of the desired $HgX$ population density is given by

$$[HgX(B)] = [HgX(B)]_{t=0} \exp(-t/\tau), \tag{2}$$

where

$$\tau^{-1} = \tau_r^{-1} + k_Q[HgX_2], \tag{3}$$

[ ] indicate particle densities, $\tau_r$ is the $HgX(B \rightarrow X)$ radiative lifetime, and $k_Q$ is the rate constant for quenching of the $HgX(B)$ state by $HgX_2$. This simple kinetic rate equation approach is analogous to that described previously in more detail for the XeF and KrF exciplex molecules.[13]

The emission waveforms were clean and decayed exponentially over two orders of magnitude of amplitude and several decay constants. The falling portion of each waveform was fit to a single exponential to determine $\tau$ for that particular $[HgX_2]$. Hence, by simply plotting $\tau$ for various $HgX_2$ vapor pressures, $k_Q$ and $\tau_r$ could be determined.

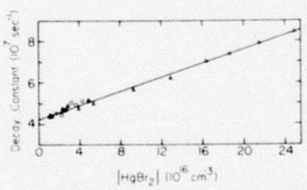



FIG. 2. Dependence of the exponential decay constant $\tau$ on $[HgBr_2]$. Data with (○) and without (Δ) 100 Torr of helium in the $HgBr_2$ cell are shown. The zero-density intercept yields the HgBr($B$) state radiative lifetime of 23.7 ns.

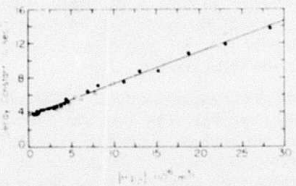



FIG. 3. Variation of $\tau$ with $[HgI_2]$ for photolysis of $HgI_2$ + 100 Torr He at 222 and 193 nm. The KrCl data (○) gives the HgI($B$) state lifetime as 27.3 ns, whereas ArF photodissociation (●) shows a longer apparent lifetime due to a higher HgI($B$) vibrational distribution. A representative data point with no helium and using KrCl photolysis (Δ) is also shown.

Figure 2 shows the dependence of $\tau$ on $[HgBr_2]$, where the solid line represents the least-squares fit of Eq. (3) to the data. The triangles shown in Fig. 2 represent data taken with no He in the absorption cell, whereas the open circles represent data with 100 Torr of helium in the tube. Although not shown, the $HgBr_2$ + He data closely followed, for all values of $[HgBr_2]$, that measured with just $HgBr_2$ in the tube, indicating that, as expected, only low HgBr($B$) vibrational levels are produced by ArF photolysis of $HgBr_2$. The zero-density intercept of the least-squares line gives the HgBr($B$) lifetime as $23.7 \pm 1.5$ ns, which is in excellent agreement with the results obtained by Djeu and Mazza[14] using a laser-induced fluorescence technique.

As mentioned previously, the $HgI_2$ wavelength threshold for photodissociation is ~230 nm, and thus lower HgI($B$) vibrational levels are created when the salt is photolyzed with KrCl radiation. Data using both KrCl (●) and ArF (○) to photodissociate $HgI_2$ (in the presence of 100 Torr He) are shown in Fig. 3, where the ArF data leads to a lifetime of 36.2 ns and a $k_Q$ of $4.5 \times 10^{-10}$ cm$^3$ s$^{-1}$. However, the ArF radiation of 193 nm can excite vibrational levels ~1 eV above the $v' = 0$ level of the $B$ state, and vibrational relaxation of the various levels could affect the apparent radiative lifetime. For KrCl excitation, where only low vibrational levels are excited, the data show a radiative lifetime of 27.3 ns and a quenching rate of $3.6 \times 10^{-10}$ cm$^3$ s$^{-1}$. Evidently, then, the radiative lifetime of the HgI($B$) state rises slightly with increasing vibrational number. Finally, the fluorescent yield from HgI* using KrCl photodissociation is much higher than that obtained using ArF,

TABLE I. Summary of radiative lifetimes of the $HgX(B)$ states ($X$ = Cl, Br, and I) and rate constant for quenching of $HgX(B)$ by $HgX_2$.

| | Experimental lifetime | Theory (Ref. 7) | $k_Q$ (cm$^3$ s$^{-1}$) |
|---|---|---|---|
| HgCl | 22.2 ns (Ref. 6) | 20 ns | $9.2 \times 10^{-11}$ (Ref. 6) |
| HgBr | 23.7 ns<br>23.2 ns (Ref. 14) | 16 | $1.7 \times 10^{-10}$ |
| HgI | 27.3 ns | 12 | $3.6 \times 10^{-10}$ |

which is also consistent with Maya's absorption measurements. Again, with no He in the $HgI_2$ tube and using KrCl photolysis, the data (Δ, only one point shown for clarity) follows the solid circles.

Table I summarizes the experimental and theoretical data for HgCl, HgBr, and HgI. The theoretical results of Duzy and Hyman, while close to the experimental value for HgCl, predicted shorter lifetimes for the heavier halides, which contrasts with these observations. Duzy and Hyman's neglect of spin orbit effects, which become increasingly important for Br and I, may contribute to the discrepancy. The increased quenching seen from the heavier salt vapors also may account for the poor performance seen by Burnham[5] for the $HgI_2$ dissociation laser.

The lifetimes and quenching rates measured here should be useful for prediction of mercury halide laser performance and design of higher-power systems. Further measurements of mercury halide quenching by other gases important to these lasers are in progress and will be reported later.

The authors gratefully acknowledge the technical assistance of D. Epp and helpful discussions with R. Burnham.

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# Quantum Electronics Letters

## Spontaneous Emission Lifetime of the $C \longrightarrow A$ Band of the XeF Molecule

R. W. WAYNANT AND J. G. EDEN

*Abstract*—The radiative lifetime of the blue $C \rightarrow A$ transition of XeF has been measured to be 93 ± 5 ns by monitoring the exponentially decaying 460 nm emission following rapid electron impact dissociation of $XeF_2$. By varying the $XeF_2$ partial pressure, the rate constant for quenching of XeF ($C$) by $XeF_2$ was determined to be $(1.8 \pm 0.5) \cdot 10^{-10}$ cm$^3$ s$^{-1}$. Finally, knowledge of the XeF $B$ and $C$ state lifetimes yields a revised value of 610 · 60 cm$^{-1}$ for the $B$ to $C$ state energy defect.

THE existence of a "broad, smooth continuum in XeF centered near 450 nm," [1] first reported in 1975, has received little attention due to the band's low intensity and to the uncertainty of its identity. Recently, however, this continuum was shown to originate from the $C$ state [2] of XeF [3] and, although theoretical calculations suggest that the $B$ and $C$ states are adjacent to one another in energy and intersect at small internuclear radii [4], the experimental results of Kligler and co-workers [5] and Brashears and Setser [6] show that the $C$ state (presumably at its equilibrium radius $R_e$) lies ~700 cm$^{-1}$ below the $B$ level.

Several XeF laser experiments conducted in the last year have suggested that the proximity of the $B$ and $C$ states in energy may have a profound effect on the laser's efficiency due to coupling of the molecular levels by thermal gas collisions. The pronounced improvement of the XeF (350 nm) laser output energy that has been observed [7] when the laser gas medium was heated to 150-200°C ($kT \sim 300$ cm$^{-1}$) raises the interesting possibility that the XeF ($C$) state may be acting as an energy reservoir to feed the lasing $B \rightarrow X$ transition. The imperative then, as several investigators have noted [5], [6], of including the $C$ state in future modeling calculations underscores the necessity of determining the $C$ state lifetime. Since only estimates were previously available, this letter will describe the measurement of the XeF ($C \rightarrow A$) radiative lifetime which has been found to be 93 ± 5 ns.

A partial energy level diagram for the XeF molecule is shown in Fig. 1. Although the potential curves are basically those given by Dunning and Hay [4]: 1) the $X$ state has been modified to reflect the ~1200 cm$^{-1}$ binding and the equilibrium radius ($R_e \simeq 2.2$ Å) experimentally determined by Smith and Kobrinsky [8]; 2) the energy defect between the $B$ and $C$ states has been adjusted to conform to [5], [6]; and 3) $R_e$ for the $B$ state has been set at 2.5 Å [8]. As shown in the figure, the $B \rightarrow X$ transition of XeF is responsible for the well-known structured band in the ultraviolet at 350 nm. The importance of this band as a high energy laser transition prompted the recent measurements [9]-[11] of the XeF ($B \rightarrow X$) radiative lifetime.

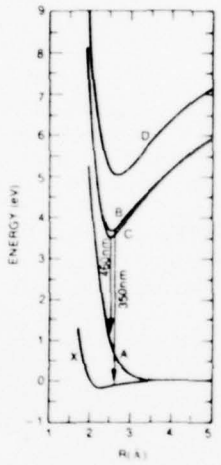



Fig. 1. Partial energy level diagram of XeF schematically showing the $C \rightarrow A$ (blue) and $B \rightarrow X$ (ultraviolet) bands (after Dunning and Hay; see [4] and text).

For the experiments described below, dissociative excitation of $XeF_2$ by electron impact was used to populate the XeF ($C$) state and subsequently determine the $C \rightarrow A$ lifetime. Pulsed ultraviolet photolysis [11] of $XeF_2$ creates too few $C$ state molecules for reliable spontaneous emission measurements to be made. Also, the narrow [~3 ns full width at half maximum (FWHM)] electron beam excitation pulse used in these experiments allows the $C$ state lifetime to be accurately determined.

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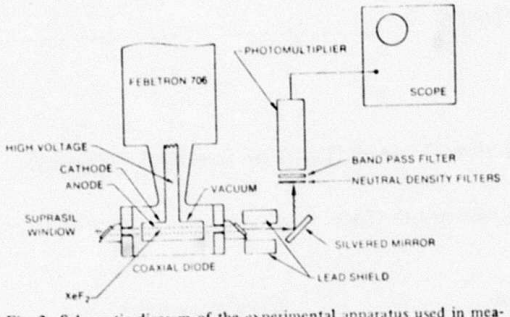



Fig. 2. Schematic diagram of the experimental apparatus used in measuring the XeF ($C \rightarrow A$) radiative lifetime.

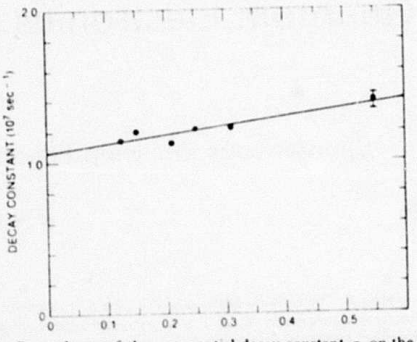



Fig. 3. Dependence of the exponential decay constant, $\tau$, on the $XeF_2$ vapor pressure. The zero pressure intercept of the least squares fitting to the data (—) gives the XeF ($C \rightarrow A$) lifetime as 93 ns. Also, the slope of the solid line yields $k_Q = 1.8 \cdot 10^{-10}$ cm$^3 \cdot$ s$^{-1}$. The one error bar shown typifies the estimated uncertainty of ~3 percent per data point.

The experimental apparatus is shown schematically in Fig. 2. A 600 kV, 3 ns FWHM voltage pulse was applied to a coaxial diode by a Febetron 706 generator. Electrons emitted from the cathode penetrated the cylindrical anode (aluminum tubing: 25 $\mu$m wall thickness, 9 mm diameter, and ~25 cm length) which contained ~0.1–0.5 torr of $XeF_2$. Upon entering the interior of the anode, the electrons dissociatively excited the $XeF_2$, forming a variety of XeF excited molecules:

$$e + XeF_2 \longrightarrow XeF\,(B, C, D) + F + e. \quad (1)$$

Limiting our attention to the $C$ state, XeF ($C$) molecules subsequently radiate in the vicinity of 460 nm. Therefore

$$XeF\,(C) \xrightarrow{\tau_r^{-1}} XeF\,(A) + h\nu\,(460\text{ nm}). \quad (2)$$

Since the $C$ state is populated in a period of time that is short compared to its esitmated lifetime, then $\tau_r$ may be determined by simply observing the exponential decay of the $C$ state molecular population. This decay is, of course, reflected in the temporal behavior of the $C \rightarrow A$ emission which was detected by an EMI 9818 QB photomultiplier. To minimize the background noise reaching the detector, a silvered mirror was used to deflect the XeF* fluorescence into the phototube. Also, lead shielding was situated around the electron diode and detector. A blue bandpass filter ($T_{max}$ = 35 percent at $\lambda_0$ = 462 nm; $\Delta\lambda$ = 50 nm) selected the desired special region and neutral density filters attenuated the signal to avoid saturation of the photomultiplier. The XeF ($C \rightarrow A$) spontaneous emission waveforms were displayed on an oscilloscope, manually digitized, and then plotted on semi-log scales by a computer. Since each flourescence waveform decayed exponentially over roughly two orders of magnitude, the exponential decay constant $\tau$ could be accurately measured for a particular $XeF_2$ concentration. By determining $\tau$ for several $XeF_2$ pressures in the range 0.1–0.5 torr, Fig. 3 was obtained which presents the dependence of $\tau$ on $XeF_2$ vapor pressure.

A simple rate equation analysis of processes by which $C$ state molecules are destroyed reveals that

$$\tau^{-1} = \tau_r^{-1} + k_Q\,[XeF_2] \quad (3)$$

where $\tau_r$ is the $C \rightarrow A$ radiative lifetime, [] denote particle densities (in units of cm$^{-3}$) and $k_Q$ is the rate constant for quenching of XeF ($C$) state molecules by $XeF_2$. From the least squares fitting of the data of Fig. 3 (denoted by the solid line), $\tau_r = 93 \pm 5$ ns and $k_Q = (1.8 \pm 0.5) \cdot 10^{-10}$ cm$^3 \cdot$ s$^{-1}$. This value for the radiative lifetime is in good agreement with that measured recently by Center (80 ns) [12] and the theoretical value of 113 ns given by Dunning and Hay [4].

As mentioned previously, low $XeF_2$ pressures were used in these experiments and buffer gas was avoided to minimize collisional coupling of the XeF $B$ and $C$ states. As a result, however, the XeF ($C$) state is not vibrationally equilibrated and therefore the radiative lifetime of 93 ns likely represents contributions from several $C$ state vibrational levels.

Kligler and co-workers [5] have recently estimated the energy separation between the XeF $C$ and $B$ levels by comparing the relative intensities of the $e$-beam excited $B \rightarrow X$ and $C \rightarrow A$ transitions at high pressure. The defect that they calculate, 700 ± 70 cm$^{-1}$, assumes 1) collisional equilibrium between the $C$ and $B$ states and 2) $(\tau_{B\rightarrow X}/\tau_{C\rightarrow A}) \simeq (1/10)$. Using the $B \rightarrow X$ lifetime of 14.2 ns [11] and the $C \rightarrow A$ value presented above, then $(\tau_{B\rightarrow X}/\tau_{C\rightarrow A}) \simeq (1/6.5)$ and so $\Delta E\,(B - C) = 610 \pm 60$ cm$^{-1}$ = 0.076 ± 0.007 eV which is in excellent agreement with the separation of 0.08 eV measured by Brashears and Setser [6].

As mentioned earlier, the ~600 cm$^{-1}$ defect between the $C$ and $B$ states may make the relatively long lived (~90 ns) $C$ state ideal as an energy storage level. If the XeF laser gas mixture is heated to ~200°C, then during lasing the close collisional coupling between the $C$ and $B$ states will drive the mixing reactions toward the depleted $B$ state (example: XeF ($B$) + Xe $\rightleftarrows$ XeF ($C$) + Xe + 600 cm$^{-1}$), thus feeding the lasing $B \rightarrow X$ transition. More complete studies of the collisional properties of the $C$ state will be required to assess the role of this level as an energy reservoir for the XeF UV laser and possibly other rare-gas halide molecules.

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The authors are grateful for the technical assistance of D. Epp and interesting discussions with R. Burnham, L. Champagne, and J. Hsia.

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